

Then with rule INIT,UNITARY and FRAME, we continue the proof

$$\begin{aligned}
 & \dots\dots \\
 & \implies \{q \mapsto |\lambda\rangle * q_1, q_2 \mapsto I\} \\
 & [q_1, q_2] := |0\rangle; \\
 & \implies \{q \mapsto |\lambda\rangle * q_1, q_2 \mapsto |00\rangle\} \\
 & X[q_1]; H[q_1]; \\
 & \implies \{q \mapsto |\lambda\rangle * q_1, q_2 \mapsto |0-\rangle\}
 \end{aligned}$$

Next, we need to prove that

$$q \mapsto |\lambda\rangle * q_1, q_2 \mapsto |0-\rangle \models (q_1, q_2 \hookrightarrow I) \wedge (q_1, q_2 \hookrightarrow P_1 \rightsquigarrow \varphi) \wedge (q_1, q_2 \hookrightarrow P_0 \rightsquigarrow \psi)$$

Each conjunction clause is proven separately. With sound rules in Figure 7 it can be derived that

$$\frac{q \mapsto |\lambda\rangle \models \top \quad q_1, q_2 \mapsto |0-\rangle \models q_1, q_2 \mapsto I}{q \mapsto |\lambda\rangle * q_1, q_2 \mapsto |0-\rangle \models \top * q_1, q_2 \mapsto I}$$

Next, because $|0-\rangle\langle 0-| \subseteq P_1$, therefore

$$(q \mapsto |\lambda\rangle * q_1, q_2 \mapsto |0-\rangle) \wedge (q_1, q_2 \hookrightarrow P_1) \equiv q \mapsto |\lambda\rangle * q_1, q_2 \mapsto |0-\rangle \models q_1, q_2, q \mapsto I \otimes |\lambda\rangle\langle\lambda| = \varphi$$

So with introduction rule for Sasaki hook, we conclude that

$$q \mapsto |\lambda\rangle * q_1, q_2 \mapsto |0-\rangle \models (q_1, q_2 \hookrightarrow P_1) \rightsquigarrow \varphi$$

Similarly, since $|0-\rangle\langle 0-| \wedge P_0 = \mathbf{0}$, thus

$$(q \mapsto |\lambda\rangle * q_1, q_2 \mapsto |0-\rangle) \wedge (q_1, q_2 \hookrightarrow P_0) \equiv \perp \models \psi$$

So we conclude that

$$q \mapsto |\lambda\rangle * q_1, q_2 \mapsto |0-\rangle \models (q_1, q_2 \hookrightarrow P_0) \rightsquigarrow \psi$$

Loop body. With rule WHILE, we prove for the loop body

$$\begin{aligned}
 & \{q_1, q_2, q \mapsto I \otimes |\lambda\rangle\langle\lambda|\} \\
 & \text{qalloc}(q_3); \\
 & \{q_1, q_2, q \mapsto I \otimes |\lambda\rangle\langle\lambda| * q_3 \mapsto I\} \\
 & \implies \{q_1, q_2, q_3 \mapsto I * q \mapsto |\lambda\rangle\} \\
 & [q_1, q_2, q_3] := |0\rangle; \\
 & \implies \{q_1, q_2, q_3 \mapsto |0\rangle * q \mapsto |\lambda\rangle\} \\
 & H[q_1]; H[q_2]; Toffoli[q_1, q_2, q_3]; \\
 & \{q_1, q_2, q_3 \mapsto \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle)|0\rangle + |11\rangle|1\rangle * q \mapsto |\lambda\rangle\} \\
 & \implies \{q_1, q_2, q_3, q \mapsto \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle)|0\rangle|\lambda\rangle + |11\rangle|1\rangle|\lambda\rangle\} \\
 & CNOT[q_3, q]; S[q]; CNOT[q_3, q]; Z[q]; \\
 & \{q_1, q_2, q_3, q \mapsto \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle)|0\rangle(ZS|\lambda\rangle) + |11\rangle|1\rangle(ZXSX|\lambda\rangle)\} \implies \\
 & \{q_3, q_1, q_2, q \mapsto \frac{1}{2\sqrt{2}}|+\rangle[(|00\rangle + |01\rangle + |10\rangle)(ZS|\lambda\rangle) + |11\rangle(ZXSX|\lambda\rangle)] \\
 & \quad + \frac{1}{2\sqrt{2}}|-\rangle[(|00\rangle + |01\rangle + |10\rangle)(ZS|\lambda\rangle) - |11\rangle(ZXSX|\lambda\rangle)]\}
 \end{aligned}$$

Next, because $q \mapsto |v\rangle + |w\rangle \implies q \mapsto |v\rangle\langle v| + |w\rangle\langle w| \implies q \mapsto |v\rangle \vee q \mapsto |w\rangle$, thus

$$\begin{aligned}
 & \{q_3, q_1, q_2, q \mapsto \frac{1}{2\sqrt{2}}|+\rangle[(|00\rangle + |01\rangle + |10\rangle)(ZS|\lambda\rangle) + |11\rangle(ZXSX|\lambda\rangle)] \\
 & \quad + \frac{1}{2\sqrt{2}}|-\rangle[(|00\rangle + |01\rangle + |10\rangle)(ZS|\lambda\rangle) - |11\rangle(ZXSX|\lambda\rangle)]\} \implies \\
 & \{q_3, q_1, q_2, q \mapsto \frac{1}{2}|+\rangle[(|00\rangle + |01\rangle + |10\rangle)(ZS|\lambda\rangle) + |11\rangle(ZXSX|\lambda\rangle)] \vee \\
 & \quad q_3, q_1, q_2, q \mapsto \frac{1}{2}|-\rangle[(|00\rangle + |01\rangle + |10\rangle)(ZS|\lambda\rangle) - |11\rangle(ZXSX|\lambda\rangle)]\}
 \end{aligned} \tag{2}$$